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(54) **HAND-HELD POWER TOOL**

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E21B 10/44 (2006.01)

E21B 11/00 (2006.01)

(52) **U.S. Cl.**

CPC **E21B 11/005** (2013.01); **B25F 5/001** (2013.01); **B25F 5/024** (2013.01); **E21B 10/44** (2013.01)

(57)

ABSTRACT

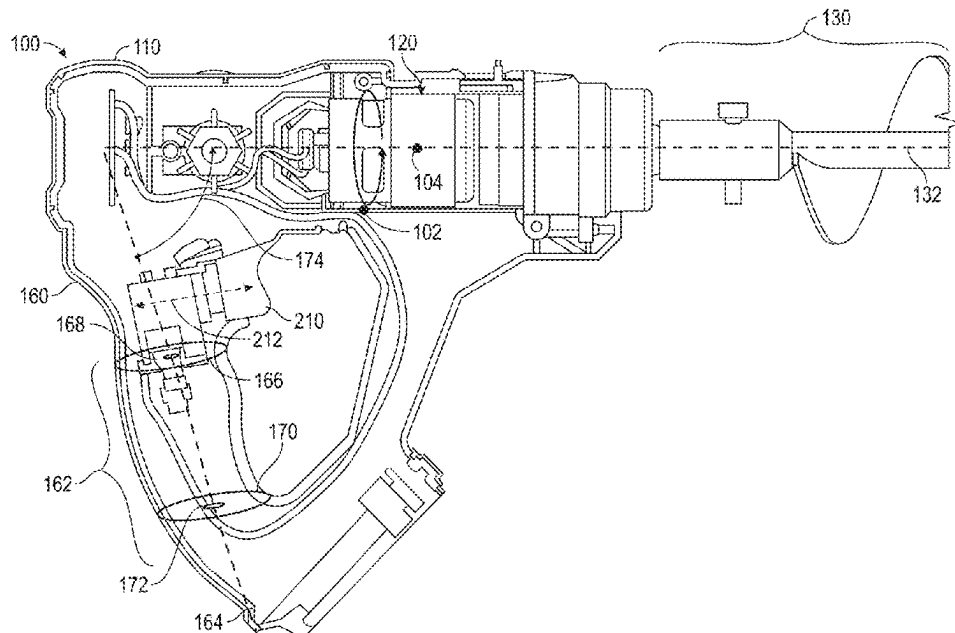
Disclosed herein is a hand-held power tool including a motor housing, a motor in the motor housing, an auger coupled to the motor and having a rotational axis, and a first handle coupled to the motor housing and including a grip. In some embodiments, the rotational axis of the auger and an axis of the grip subtend to an acute angle.

(58) **Field of Classification Search**

CPC E21B 11/005; E21B 10/44; B25F 5/001; B25F 5/024; B25F 5/008

See application file for complete search history.

18 Claims, 8 Drawing Sheets



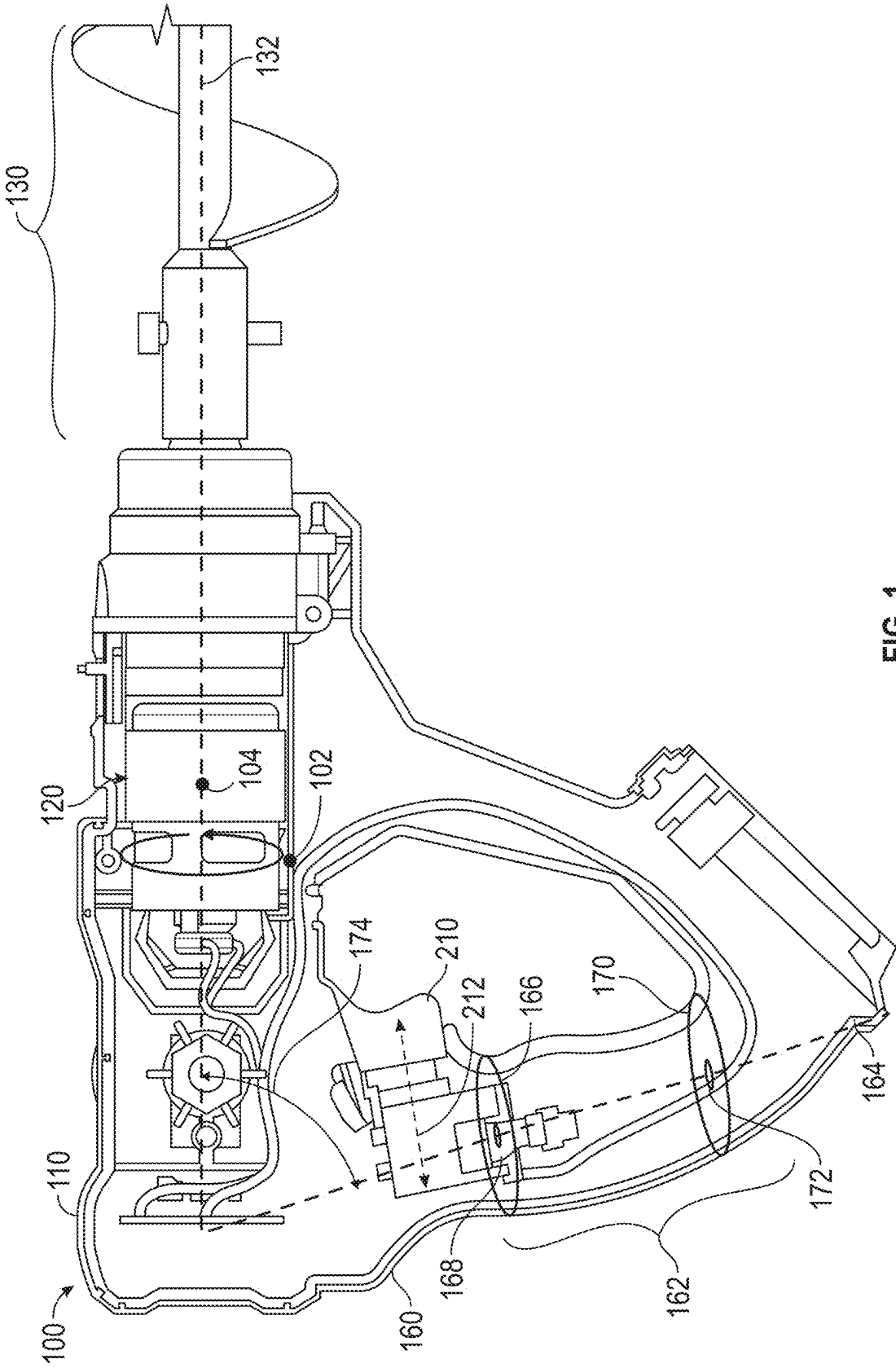


FIG. 1

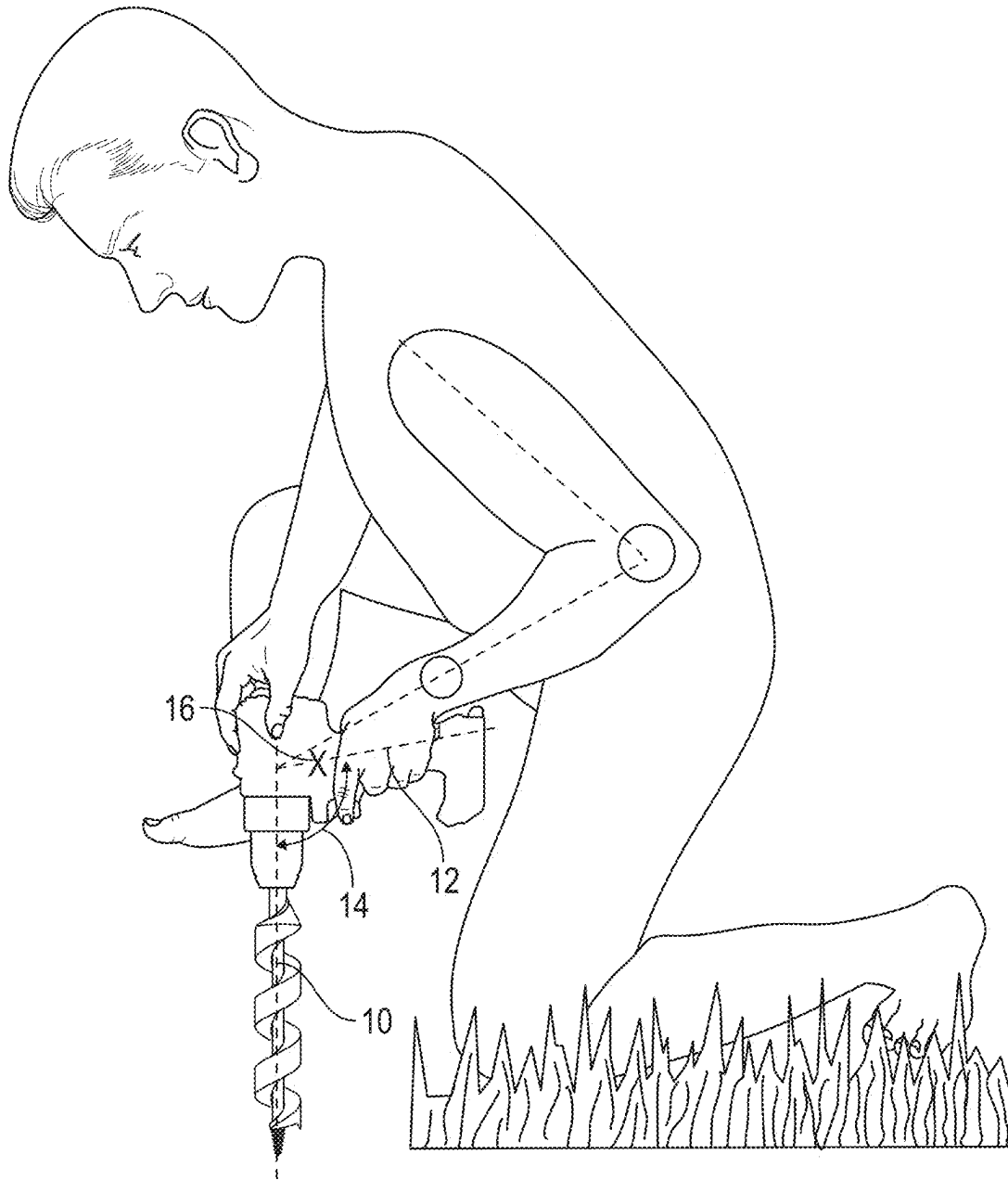


FIG. 2A
(Prior Art)

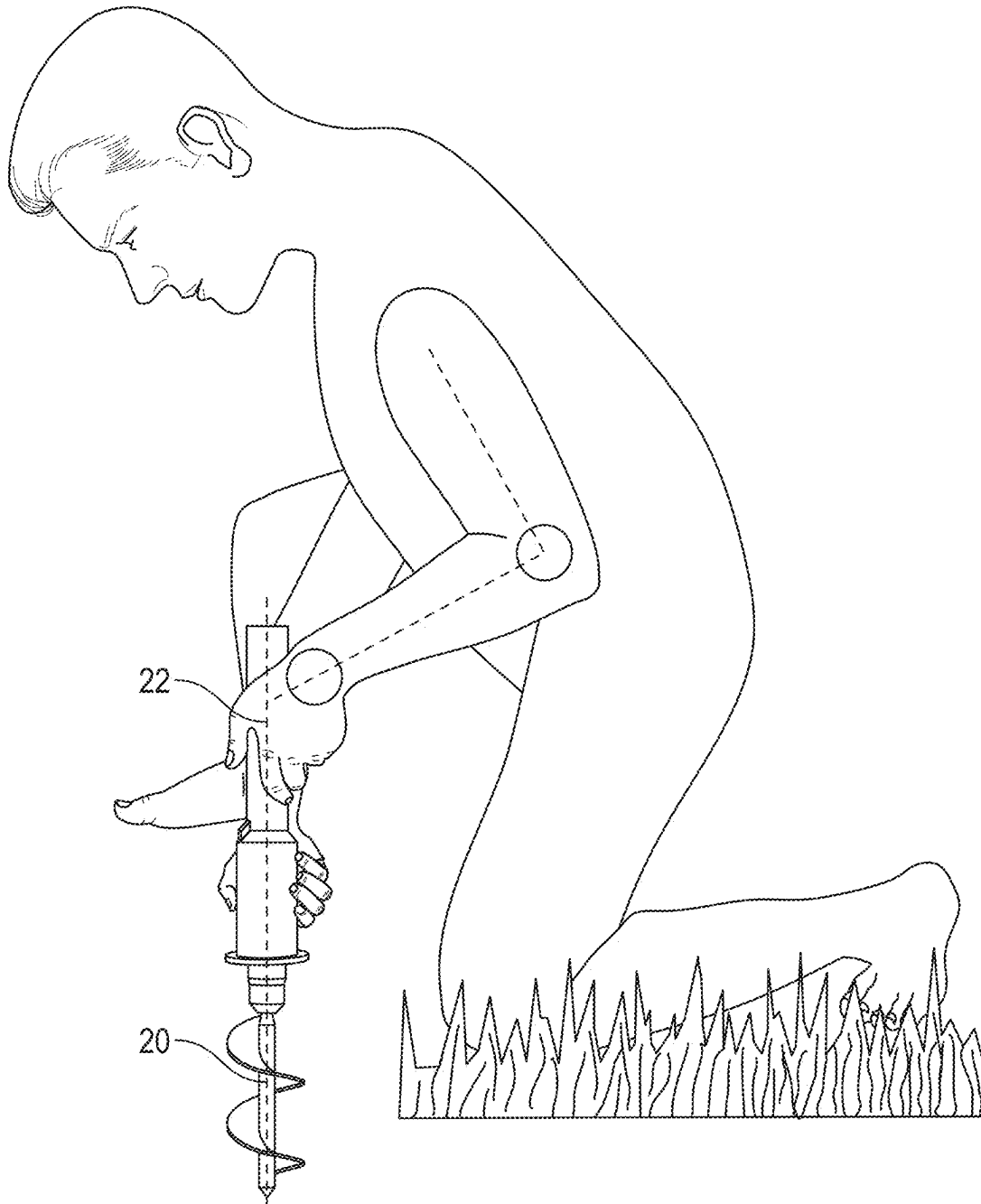


FIG. 2B
(Prior Art)

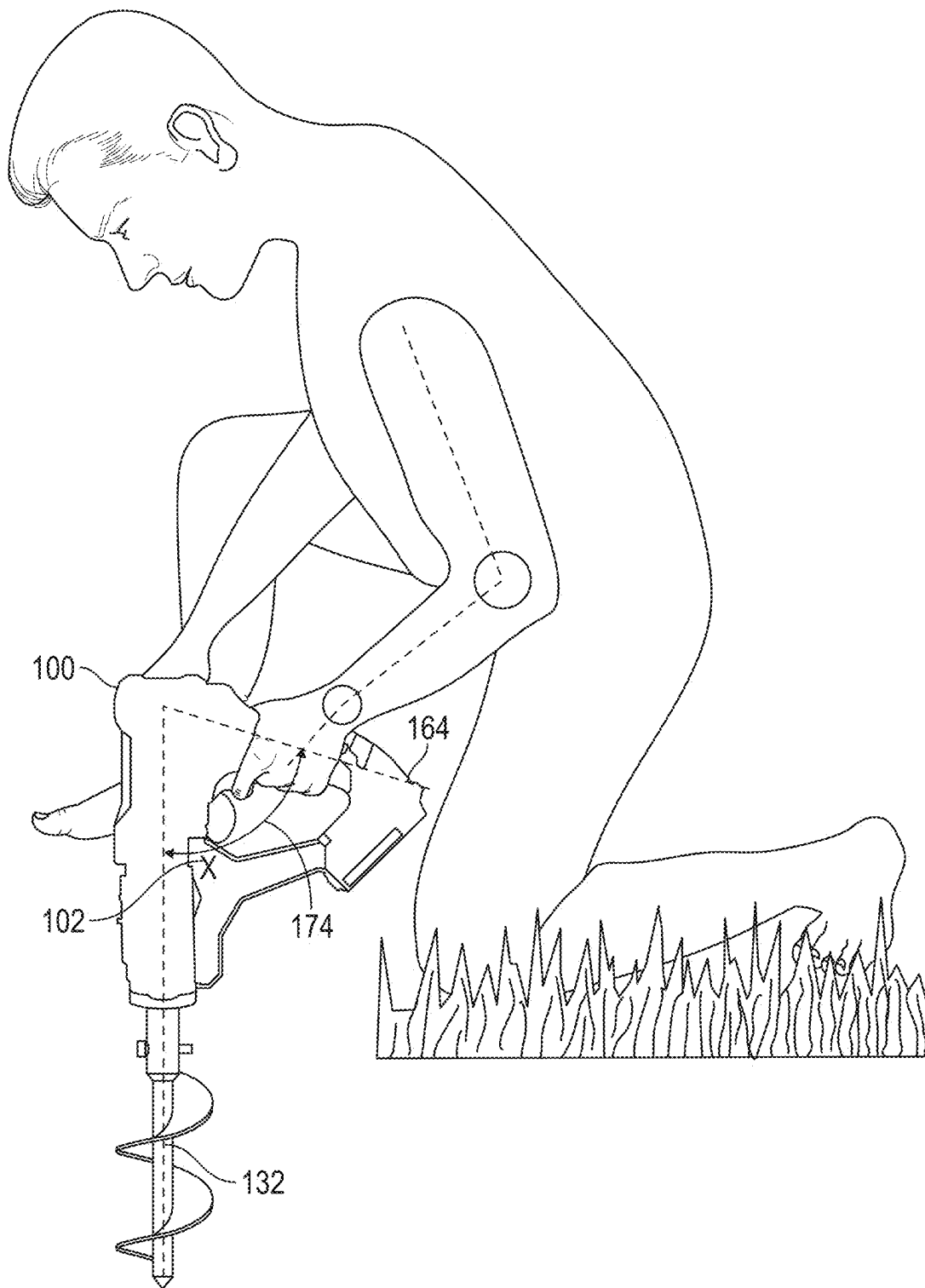


FIG. 2C

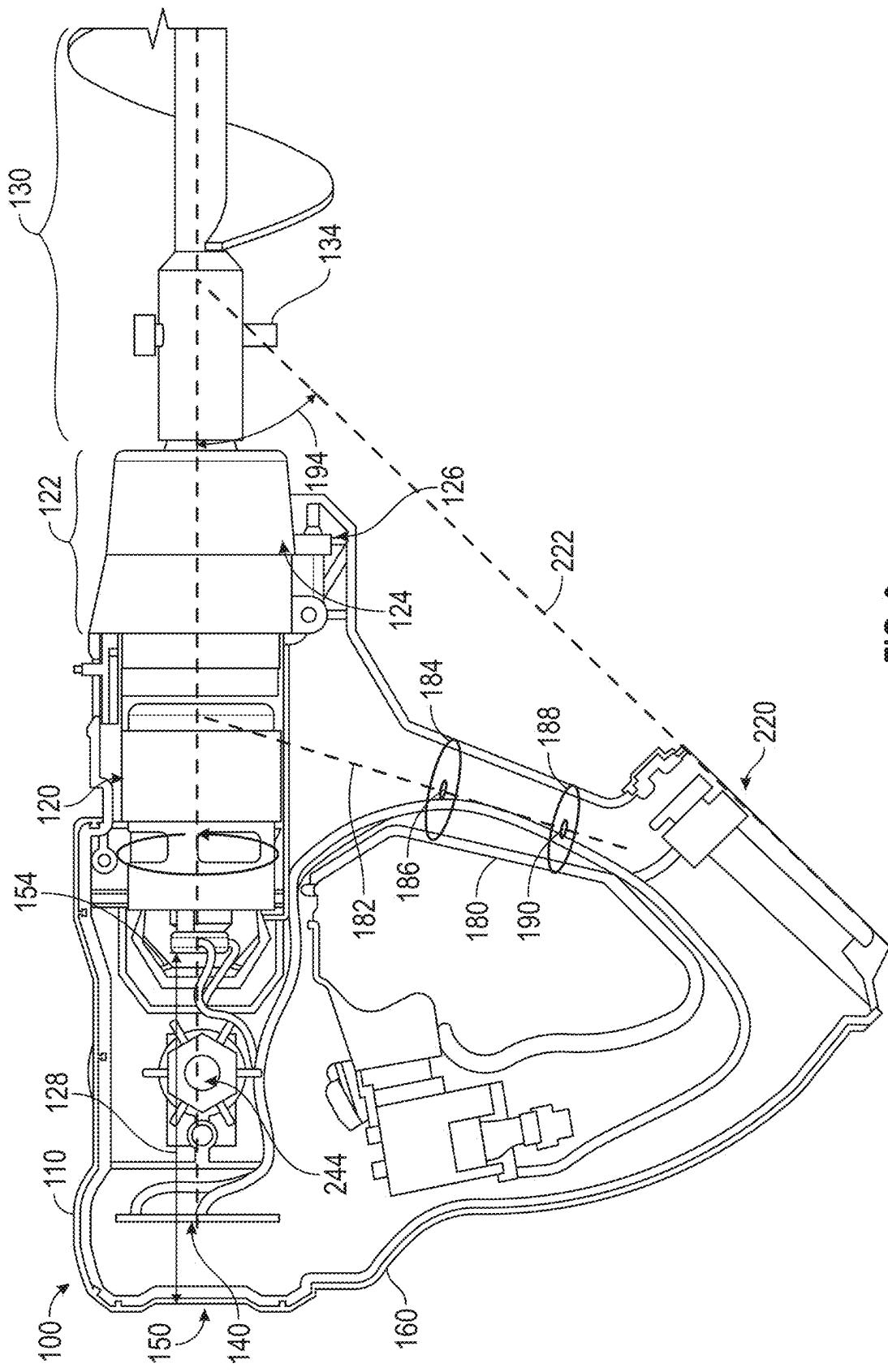


FIG. 3

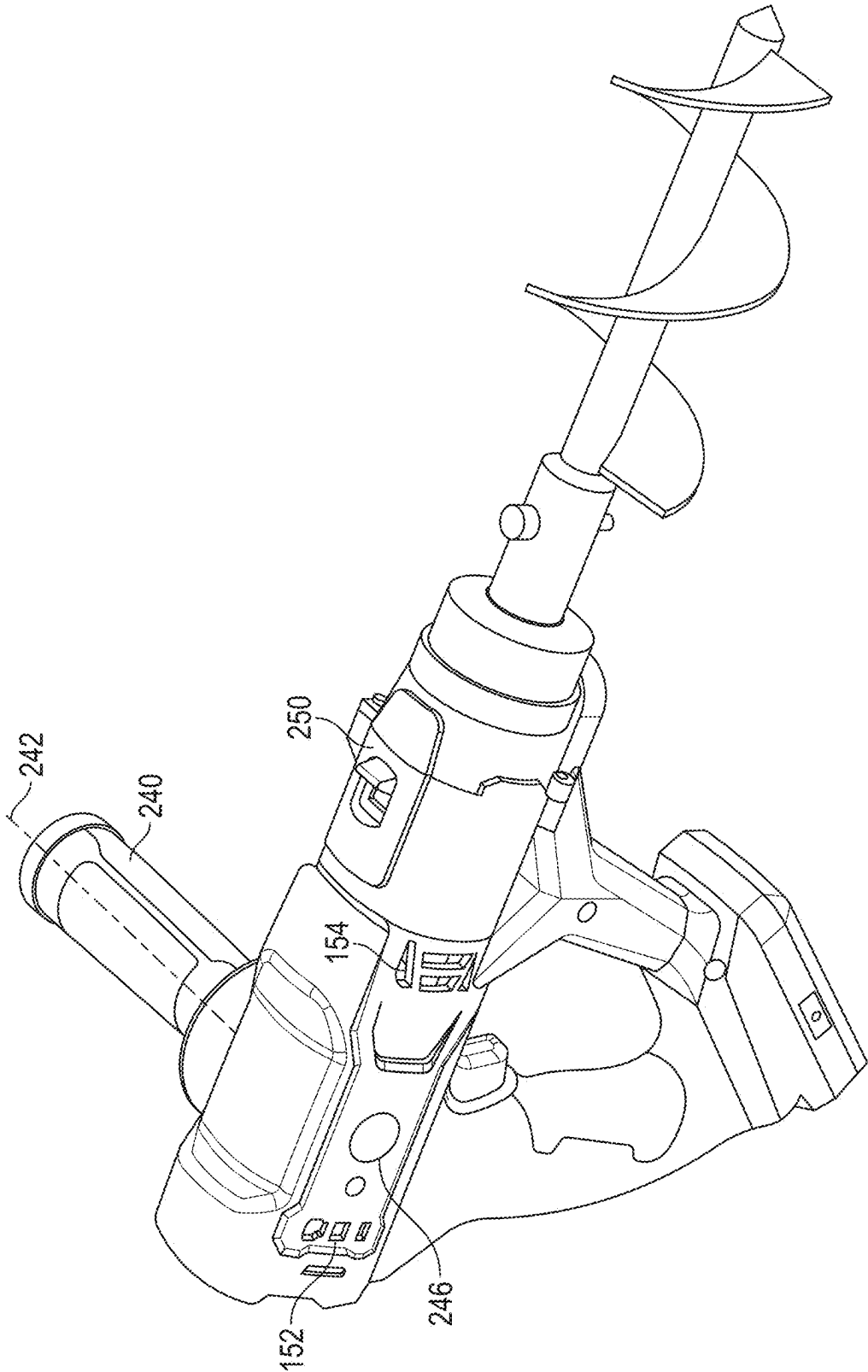


FIG. 4

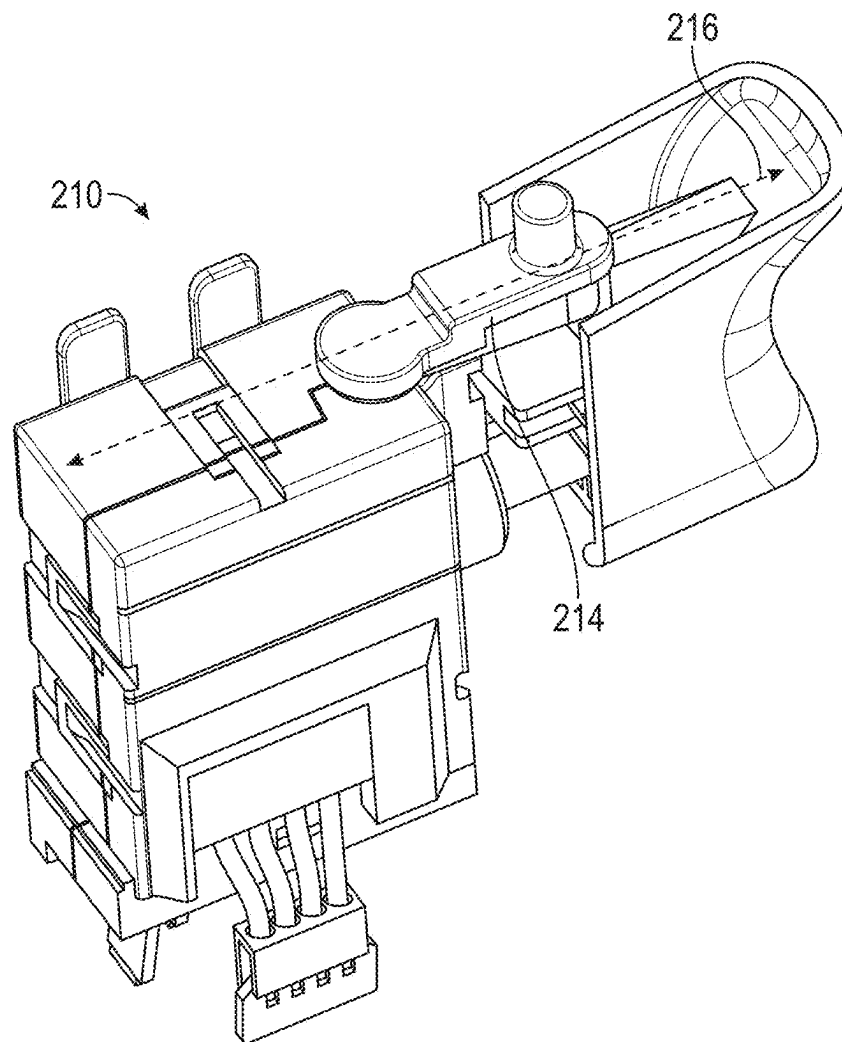


FIG. 5

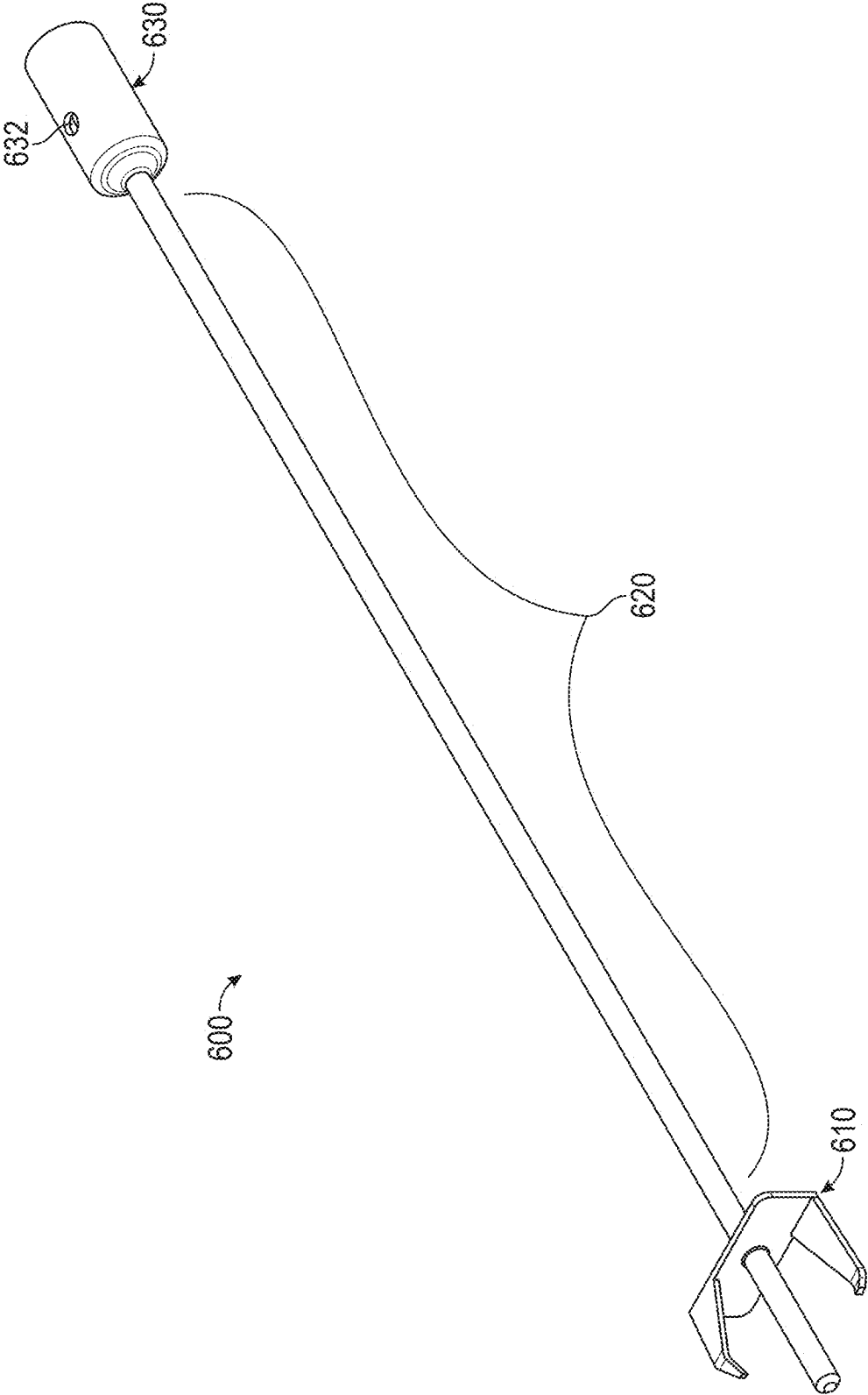


FIG. 6

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HAND-HELD POWER TOOL**FIELD**

This disclosure generally relates to a hand-held power tool. More specifically, this disclosure relates to a hand-held power tool for achieving a comfortable wrist angle during operation.

BACKGROUND

Hand tools, such as a hand auger or a screwdriver, may require significant effort to operate. Moreover, because such tools are limited by how quickly a user can operate them, hand tools may be inefficient. Power tools may solve many of these shortcomings. For example, a user operating a drill may be able to drive a screw more quickly and with less effort than would be required using a screwdriver. Power tools are useful in many contexts such as construction, metalworking, gardening, and housework. For example, power tools may be used for drilling, cutting, sanding, digging, fastening, and/or the like. Power tools are often designed for their intended application. As a result, some power tools may be poorly suited to certain applications (e.g., those outside their intended use). For example, a drill for driving fasteners may be poorly suited to dig holes.

BRIEF SUMMARY

Disclosed herein are hand-held power tools. The hand-held power tools disclosed herein may promote comfortable operation by reducing the strain on a user from operating the hand-held power tools. For example, the hand-held power tools disclosed herein promote a comfortable wrist/elbow position for a user by reducing strain induced by ulnar deviation experienced by a user when operating the hand-held power tools in a vertical orientation. As another example, the hand-held power tools disclosed herein reduce an amount of force required to hold the hand-held power tools in a vertical orientation by positioning the center of mass of the hand-held power tools nearby a tool head of the hand-held power tools. As a third example, the hand-held power tools disclosed herein reduce strain associated with a user's wrist contacting a battery pack of the hand-held power tools by positioning the battery pack to provide wrist clearance for a user.

One implementation of the present disclosure is a hand-held power tool including a motor housing, a motor in the motor housing, an auger coupled to the motor and including a rotational axis, and a first handle coupled to the motor housing and including a grip. In some embodiments, the rotational axis of the auger and an axis of the grip subtend an acute angle as measured from the auger.

In some embodiments, the acute angle is 65 to 90 degrees. In some embodiments, the motor housing includes an air intake spaced a distance from the motor. In some embodiments, the hand-held power tool includes a controller positioned between the air intake and the motor. In some embodiments, the motor is positioned proximal to the auger. In some embodiments, the motor is positioned between the auger and a center of mass of the hand-held power tool. In some embodiments, the hand-held power tool includes a clutch configured to interrupt transmission of torque from the motor to the auger when a threshold torque value is exceeded. In some embodiments, the clutch includes a selector to select the threshold torque value from a number of torque values, wherein the selector is mechanically lim-

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ited to a single selection from the number of torque values. In some embodiments, the angle is 74 degrees. In some embodiments, the hand-held power tool includes a battery foot coupled to the first handle and configured to slidably engage a battery, wherein an axis of engagement of the battery foot and the rotational axis of the auger subtend a second angle of 15 to 75 degrees measured from the first handle. In some embodiments, the hand-held power tool includes a bridge coupled to the motor housing and the battery foot. In some embodiments, the motor is positioned in the motor housing such that an axis of the bridge intersects the motor. In some embodiments, the hand-held power tool includes a second handle, wherein a longitudinal axis of the second handle is perpendicular to the rotational axis of the auger and the axis of the grip portion. In some embodiments, the motor housing includes an air intake spaced a distance from the motor. In some embodiments, a controller is positioned closer to the air intake than the motor. In some embodiments, a majority of the motor is positioned between a center of mass of the hand-held power tool and the auger. In some embodiments, the battery foot does not extend rearward of the axis of the grip.

Another implementation of the present disclosure is a hand-held power tool including a motor housing, a motor in the motor housing, a tool head coupled to the motor and including a rotational axis, a battery foot configured to slidably engage a battery, and a handle coupled to the motor housing and including a grip, wherein an axis of engagement of the battery foot and the rotational axis of the tool head subtend an angle greater than 0 degrees as measured from the handle.

In some embodiments, the angle is 15 to 75 degrees. In some embodiments, the hand-held power tool includes a bridge coupled to the motor housing and the battery foot. In some embodiments, the motor is positioned in the motor housing such that an axis of the bridge intersects the motor. In some embodiments, the hand-held power tool includes a second handle, wherein a longitudinal axis of the second handle is perpendicular to the rotational axis of the tool head and an axis of the grip. In some embodiments, the motor is positioned proximal to the tool head. In some embodiments, the tool head includes an auger bit or a weeding bit. In some embodiments, the hand-held power tool includes a second handle positioned behind the motor.

Another implementation of the present disclosure is a hand-held power tool including a motor housing, a motor in the motor housing, wherein the motor housing includes an air intake spaced a distance from the motor, a controller positioned between the air intake and the motor, a tool head coupled to the motor and including a rotational axis, and a handle coupled to the motor housing and including a grip, and wherein the motor is positioned between a center of mass of the hand-held power tool and the tool head.

In some embodiments, the hand-held power tool includes a bridge coupled to the motor housing, wherein the motor is positioned in the motor housing such that an axis of the bridge intersects the motor. In some embodiments, the motor is positioned proximal to the tool head. In some embodiments, the tool head includes an auger bit or a weeding bit.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 illustrates a side view of an exemplary hand-held power tool, in accordance with an embodiment.

FIG. 2A is an exemplary illustration of a traditional tool in use by a user.

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FIG. 2B is an exemplary illustration of another traditional tool in use by a user.

FIG. 2C is an exemplary illustration of the hand-held power tool of FIG. 1 in use by a user, in accordance with an embodiment.

FIG. 3 illustrates exemplary components of the hand-held power tool of FIG. 1, in accordance with an embodiment.

FIG. 4 illustrates a perspective view of the hand-held power tool of FIG. 1, in accordance with an embodiment.

FIG. 5 illustrates a perspective view of a trigger of the hand-held power tool of FIG. 1, in accordance with an embodiment.

FIG. 6 illustrates an exemplary tool head of the hand-held power tool of FIG. 1, in accordance with an embodiment.

DETAILED DESCRIPTION

In the following description of embodiments, reference is made to the accompanying drawings which form a part hereof, and in which it is shown by way of illustration specific embodiments which can be practiced. It is to be understood that other embodiments can be used and structural changes can be made without departing from the scope of the disclosed embodiments.

Referring generally to the FIGURES, described herein is a hand-held power tool having a handle angle that promotes a comfortable wrist and/or elbow position for a user during operation. Speaking generally, the hand-held power tool may reduce strain on a user caused by ulnar deviation when the hand-held power tool is used in a vertical orientation. Additionally or alternatively, a center of mass of the hand-held power tool may be positioned nearby a tool head of the hand-held power tool, thereby increasing a stability of the hand-held power tool when used in a vertical orientation.

FIG. 1 illustrates a side view of exemplary hand-held power tool 100. Hand-held power tool 100 includes housing 110, motor 120 housed in housing 110, work implement 130 coupled to motor 120, and first handle 160. Motor 120 is attached to work implement 130, which, as shown in FIG. 1, may include an auger. A user may grip first handle 160 to operate hand-held power tool 100. In some embodiments, an axis of first handle 160 (shown as grip axis 164) subtends an acute angle (shown as first angle 174) with a rotation axis of work implement 130 (shown as rotational axis 132) that promotes a comfortable wrist and/or elbow position for a user during operation. In this way, hand-held power tool 100 may reduce strain on a user. Traditional tools (e.g., a conventional power drill, etc.) may feature a handle angled greater than 90 degrees relative to a tool head (e.g., at a 101 degree angle relative to a tool head, etc.), thereby causing strain on a user when the tool is used in a vertical orientation (e.g., when a tool head is used to drive vertically towards the ground, etc.). Conversely, first angle 174 of hand-held power tool 100 may reduce and/or eliminate this strain by producing a neutral wrist and/or elbow position, especially in digging contexts.

First handle 160 may include grip 162. Grip 162 may correspond to a portion of first handle 160 that receives the middle, ring, and pinky fingers of a user. In various embodiments, grip 162 is bounded at the top by trigger 210 and bounded at the bottom by base unit 220. For example, a top of grip 162 may be adjacent to trigger 210 and a bottom of grip 162 may be adjacent to base unit 220. Grip 162 may include grip axis 164. Grip axis 164 may be characterized by a line passing through first center 168 of cross-section 166 and second center 172 of cross-section 170. Cross-section 166 and cross-section 170 may be taken with respect to an

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outside circumference of grip 162. Cross-section 166 may be positioned adjacent to trigger 210 (e.g., where a middle finger of a user would contact first handle 160, etc.). Cross-section 170 may be positioned adjacent to base unit 220 (e.g., where a pinky finger of a user would contact first handle 160, etc.). In various embodiments, cross-section 166 and/or cross-section 170 are taken with respect to a plane parallel to an axis of actuation of trigger 210 (shown as axis 212). Many planes parallel to axis 212 may exist. In various embodiments, cross-section 166 and/or cross-section 170 are taken with respect to the specific plane which (i) has a normal vector that intersects rotational axis 132 and (ii) is parallel to axis 212. As shown in FIG. 1, the cross-sectional plane of cross-section 166 and/or cross-section 170 extend into and out of the page. In some embodiments, first center 168 and second center 172 correspond to a centroid of cross-section 166 and cross-section 170 respectively. It should be understood that while first center 168 and second center 172 are shown as circular in FIG. 1, this is for illustrative purposes only and first center 168 and second center 172 are respective points in 3-dimensional space. As mentioned above, rotational axis 132 and grip axis 164 may subtend first angle 174 measured from work implement 130. First angle 174 may be in a range of 65 degrees to 90 degrees. For example, first angle 174 may be a 74-degree angle.

Turning now to FIG. 2A, an exemplary illustration of a traditional tool in use by a user is shown for purposes of comparison. Unlike hand-held power tool 100, an axis of the handle of the traditional tool (shown as grip axis 12) may subtend an obtuse angle (shown as angle 14) with respect to a rotational axis of a work implement of the traditional tool (shown as rotational axis 10). Angle 14 may cause strain on a user when operating the tool in a vertical orientation, as shown, due to abduction of the user's wrist and/or positioning of the user's arm. For example, when using the tool, a user's arm may start in a highly compressed position, thereby causing strain on the user's elbow. In contrast, first angle 174 of hand-held power tool 100 is acute which may reduce abduction of a user's wrist to create a more natural wrist position, thereby reducing strain on a user. In various embodiments, a natural wrist position is one in which a line passing through the knuckles (e.g., metacarpophalangeal joints) of an individual's hand is approximately perpendicular to an axis of the individual's forearm (e.g., running from the radial head to the ulnar fovea).

FIG. 2B is another exemplary illustration of a traditional tool in use by a user, shown for the purposes of comparison. Unlike hand-held power tool 100, an axis of the handle of the traditional tool (shown as grip axis 22) may be coaxial with a rotational axis of the traditional tool (shown as rotational axis 20), thereby causing twisting forces from operation of the tool to be transmitted to a user's wrist. These twisting forces may be uncomfortable or even dangerous. Hand-held power tool 100 may reduce this twisting force by positioning grip axis 164 separate of rotational axis 132 (e.g., as shown in FIG. 2C), thereby increasing the usability and comfort of hand-held power tool 100. In addition, hand-held power tool 100 may include second handle 240 (as discussed below) to facilitate stabilizing hand-held power tool 100 and/or reduce transmission of twisting forces to a user's wrist. Moreover, like the traditional tool in FIG. 2A, the traditional tool in FIG. 2B may cause increased abduction of a user's wrist, which may lead to strain and discomfort. However, as discussed above,

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hand-held power tool **100** may reduce this strain by creating a more natural wrist angle that involves less abduction of the user's wrist.

FIG. 2C is an exemplary illustration of hand-held power tool **100** in use by a user. In various embodiments, first angle **174** promotes a neutral wrist and/or elbow position for a user during operation of hand-held power tool **100**, especially in digging contexts. For example, first angle **174** may reduce abduction/bending of a user's wrist, thereby reducing strain on the user. In some embodiments, hand-held power tool **100** facilitates vertical operation (e.g., in digging contexts as shown in FIG. 2C). For example, center of mass **102** may be positioned near work implement **130**. Traditional tools (like that discussed in FIG. 2A) may have a weight distribution configured for horizontal operation (e.g., for securing a fastener to a wall, etc.). For example, a conventional power drill (such as the tool of FIG. 2A) may have a motor positioned towards a rear of the drill (e.g., positioned at or above a grip of the handle, etc.), thereby creating a relatively high center of mass (shown as center of mass **16** in FIG. 2A) during vertical operation. Conversely, in some embodiments herein, motor **120** may be positioned towards a front of hand-held power tool **100** (as shown in FIG. 1), thereby lowering the center of mass during vertical operation, reducing a tilting force, and increasing the usability of hand-held power tool **100**. In various embodiments, motor **120** is positioned forward of a center of mass of hand-held power tool **100** (shown as center of mass **102** in FIG. 1). For example, a center of motor **120** may be positioned forward of center of mass **102**. Additionally or alternatively, a center of mass of motor **120** (shown as center of mass **104** in FIG. 1) is positioned between center of mass **102** and work implement **130**.

Referring again to FIG. 1, first handle **160** may be coupled to housing **110**. For example, first handle **160** may be integral with housing **110**. In some embodiments, first handle **160** is positioned rearward of center of mass **102**, thereby increasing a stability of hand-held power tool **100** during use. Housing **110** may be elongated. Traditional tools may feature a short housing (e.g., as measured from a front of the tool to a rear of the tool). However, by elongating housing **110**, hand-held power tool **100** may position first handle **160** farther from a front of the tool, thereby positioning a user's hands farther from the ground during vertical operation and/or creating a more ergonomic body position that facilitates greater arm extension during operation. In various embodiments, first handle **160** extends from housing **110** along grip axis **164**.

In some embodiments, trigger **210** is coupled to first handle **160**. Trigger **210** may activate motor **120**. In various embodiments, trigger **210** is a variable speed trigger. For example, a speed of motor **120** may be varied by pulling (e.g., actuating) trigger **210** more or less. Trigger **210** may be actuated along axis **212**. For example, a user may actuate trigger **210** using an index finger to cause trigger **210** to operate a switch which in turn causes power to be provided to motor **120**. Trigger **210** is shown in greater detail with reference to FIG. 5, below.

Motor **120** may include a direct current (DC) motor. Additionally or alternatively, motor **120** may include an alternating current (AC) motor. Motor **120** may provide power to work implement **130**. For example, work implement **130** may be directly coupled to motor **120** (e.g., via a drive shaft, etc.) to receive mechanical power therefrom. Alternatively, the work implement **130** may be coupled to the motor **120** through a gear mechanism (not shown) or other transmission. In various embodiments, motor **120** is

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positioned proximal to work implement **130**. In various embodiments, motor **120** is coaxial to rotational axis **132**.

Work implement **130** may include an auger (as shown in FIG. 1). For example, work implement **130** may include an earth auger having a rotating shaft with one or more blades (e.g., helical screw blades, etc.) attached at an end of the rotating shaft. Additionally or alternatively, work implement **130** may include other tools/bits such as a weeding bit (shown in FIG. 6). In various embodiments, a user operates hand-held power tool **100** to create a hole in the ground using work implement **130** (e.g., as shown in FIG. 2). In various embodiments, work implement **130** is constructed of a metal. However, it should be understood that work implement **130** may be constructed of other materials (or composites thereof) as well. In various embodiments, work implement **130** is changeable. For example, hand-held power tool **100** may include a coupling mechanism (shown as bolt **134**) to mechanically couple one or more tool heads to motor **120** (e.g., via a chuck, a pin linkage, etc.). In various embodiments, work implement **130** includes a receptacle that receives a driven shaft of motor **120**. For example, work implement **130** may slide over a driven shaft of motor **120** and be coupled to the driven shaft via bolt **134** such that work implement **130** may not slide off of the driven shaft and may not rotate relative to the driven shaft. Bolt **134** may pass through a hole in work implement **130** and/or a driven shaft of motor **120**. In some embodiments, bolt **134** is secured via a fastener such as a cotter pin. Additionally or alternatively, bolt **134** may screw into work implement **130** and/or a driven shaft of motor **120** and be secured thereby. In various embodiments, work implement **130** rotates about rotational axis **132**.

FIG. 3 illustrates exemplary components of hand-held power tool **100**. Housing **110** may include air intake **150**. In some embodiments, housing **110** includes air intake **152** (shown in FIG. 4) and air exhaust **154**. Air intake **150**, air intake **152**, and/or air exhaust **154** may include one or more apertures in housing **110** to facilitate a transfer of air between an outside of housing **110** and an inside of housing **110**. For example, air may be drawn into housing **110** via air intake **150** and/or air intake **152**, pass through motor **120**, and exit housing **110** via air exhaust **154**. In various embodiments, hand-held power tool **100** includes a fan. For example, an electric fan may be coupled next to air intake **150** to dissipate heat from motor **120**. Additionally or alternatively, motor **120** may include a fan. In various embodiments, a portion of hand-held power tool **100** nearby work implement **130** is considered a "front" of hand-held power tool **100** and a portion of hand-held power tool **100** nearby air intake **150** is considered a "back" of hand-held power tool **100**.

In various embodiments, motor **120** is spaced distance **128** apart from air intake **150**. Spacing motor **120** distance **128** apart from air intake **150** may provide various benefits. For example, moving motor **120** forward (e.g., towards work implement **130**, etc.) may create space in housing **110** to position additional components, such as controller **140**. As another example, moving motor **120** forward may facilitate increased airflow from air intake **150** to an inside of housing **110**, thereby improving heat transfer for components such as motor **120**. Furthermore, moving the motor **120** forward lowers the center of mass of the tool when used in a vertical orientation as explained above. In some embodiments, distance **128** is at least 20 millimeters (mm). Additionally or alternatively, distance **128** may be greater than a length of motor **120**. For example, if motor **120** is 80 mm along its axis, then distance **128** may be 85 mm.

Hand-held power tool **100** may include controller **140**. Controller **140** may be positioned within housing **110**. In various embodiments, controller **140** is positioned between air intake **150** and motor **120**. Controller **140** may include a circuit that receives inputs from trigger **210** and/or base unit **220** and controls motor **120** based on the inputs. For example, controller **140** may control a speed of motor **120** based on an amount of depression of trigger **210**.

Motor **120** may include clutch **122**. Clutch **122** may engage/disengage mechanical power transmission from a drive shaft of motor **120** to a driven shaft (e.g., a drill chuck, work implement **130**, etc.). Clutch **122** may include selector **124**. Selector **124** may facilitate selection of a threshold torque value from a number of torque values. For example, clutch **122** may interrupt transmission of torque from a shaft of motor **120** to a driven shaft (e.g., a drill chuck, work implement **130**, etc.) when a threshold torque value is exceeded, and a user may select the threshold torque value using selector **124**. In some embodiments, clutch **122** includes a torque limiter (shown as limiter **126**). Limiter **126** may mechanically limit selector **124** to a single selection from a number of torque values. In some embodiments, limiter **126** includes a mechanical fastener. For example, limiter **126** may include a mechanical fastener that fastens a clutch collar of selector **124** to housing **110**. Additionally or alternatively, limiter **126** may include a shear pin, a synchronous magnetic torque limiter, a ball detent type limiter, a pawl and spring type limiter, and/or the like. In various embodiments, selector **124** is mechanically limited to a single threshold torque value via limiter **126**. In some embodiments, motor **120** includes an impact driver. The impact driver may facilitate digging through obstacles in the ground such as rocks or roots and/or digging through hard soil.

Hand-held power tool **100** may include base unit **220**. In various embodiments, base unit **220** is coupled to first handle **160**. Base unit **220** may receive a power source to power hand-held power tool **100**. For example, base unit **220** may include a battery footing to receive a slide-type battery pack. As another example, base unit **220** may include a corded base unit to receive AC power. Base unit **220** may slidably engage a battery along axis of engagement **222**. For example, a user may slide a battery onto a battery footing of base unit **220** along axis of engagement **222**. In various embodiments, a battery pack is slid onto base unit **220** from behind (e.g., such that the battery pack travels along axis of engagement **222** from first handle **160** towards work implement **130**).

In some embodiments, base unit **220** and/or the power source (e.g., a slide-type battery pack, etc.) include a release mechanism. For example, a slide-type battery pack may include a button that releases a mechanical linkage holding the battery pack in place such that the battery pack can be slid off base unit **220**. Rotational axis **132** and axis of engagement **222** may subtend second angle **194** measured from first handle **160**. In various embodiments, second angle **194** is in a range of 30 degrees to 55 degrees. In some embodiments, second angle **194** is in a range of 0 degrees to 90 degrees. For example, second angle **194** may range from 15 degrees to 75 degrees. In various embodiments, second angle **194** promotes comfortable operation of hand-held power tool **100** by preventing wrist and/or forearm contact with base unit **220** and/or the power source during vertical operation of hand-held power tool **100**. For example, second angle **194** may provide wrist clearance when operating hand-held power tool **100** vertically such that a user's wrist is clear of (e.g., does not contact under normal operation) the

battery pack, thereby preventing strain/discomfort on the user. In various embodiments, hand-held power tool **100** achieves this wrist clearance by positioning base unit **220** so that it does not extend beyond grip axis **164**. In comparison, traditional tools may include a battery pack that extends beyond a grip axis of the tool such that the battery contacts a user's wrist when the user abducts/adducts their wrist during normal operation (e.g., as shown in FIG. 2C). Additionally, second angle **194** may facilitate insertion/ejection of the power source from base unit **220**.

Hand-held power tool **100** may include bridge **180**. Bridge **180** may be coupled to housing **110**. For example, bridge **180** may be integral with housing **110**. In various embodiments, bridge **180** mechanically supports first handle **160**. For example, bridge **180** may prevent cantilevering of first handle **160** during operation of hand-held power tool **100**. In various embodiments, bridge **180** is coupled to base unit **220**. For example, bridge **180** may extend between housing **110** and a battery foot of base unit **220** such that bridge **180** is coupled to housing **110** and the battery foot of base unit **220**. In some embodiments, bridge **180** extends from housing **110** along axis **182**. Axis **182** may be characterized by a line passing through first center **186** of cross-section **184** and second center **190** of cross-section **188**. Cross-section **184** and cross-section **188** may be taken with respect to an outside circumference of bridge **180**. In some embodiments, first center **186** and second center **190** correspond to a centroid of cross-section **184** and cross-section **188** respectively. Cross-section **184** may be positioned adjacent to housing **110**. Cross-section **188** may be positioned adjacent to base unit **220**. In various embodiments, motor **120** is positioned such that axis **182** intersects motor **120**.

FIG. 4 illustrates a perspective view of exemplary hand-held power tool **100**. Hand-held power tool **100** may include second handle **240**. Second handle **240** facilitates stabilizing hand-held power tool **100**. For example, a user may use second handle **240** to drive work implement **130** into the ground while preventing hand-held power tool **100** from twisting when it encounters obstacles such as rocks. Second handle **240** may be coupled to housing **110**. For example, second handle **240** may be integral with housing **110**. In some embodiments, second handle **240** is detachably coupled to housing **110** via a fastener (shown as fastener **244** in FIG. 3). For example, housing **110** may include an attachment point (shown as attachment point **246**) having a female thread that receives a threaded portion on an end of second handle **240**. In various embodiments, attachment point **246** is positioned near a rear of hand-held power tool **100**. Traditional tools may position a secondary handle near a front of the tool (e.g., near the chuck, etc.). However, this may limit a range of motion of the user because the user's hands start on different planes, thereby causing one arm to be more extended than another during use. By positioning attachment point **246** near a rear of hand-held power tool **100**, a user's hands start on a similar plane, thereby facilitating a greater range of motion. In some embodiments, second handle **240** extends from housing **110** along axis **242**. Axis **242** is the longitudinal axis of second handle **240**. Axis **242** may be perpendicular to rotational axis **132** and grip axis **164**. A user may grip second handle **240** to operate hand-held power tool **100**. For example, a user may grip first handle **160** with a first hand and second handle **240** with a second hand. In various embodiments, second handle **240** reduces torquing fatigue on a user's hand/wrist/arm. For example, a user may use a first hand to operate hand-held power tool **100** via grip **162** and may experience a rotational force (e.g., about rotational axis **132**) due to the drilling

action of work implement **130**. This rotational force may be especially pronounced when work implement **130** encounters an obstacle such as a rock, thereby causing a discomfort on a user's hand/wrist/arm. However, a user may hold second handle **240** using a second hand to further stabilize hand-held power tool **100**, thereby reducing strain on their other hand/wrist/arm. Second handle **240** is shown on a left side of hand-held power tool **100** (when viewed from behind) in FIG. **4**. However, it should be understood that second handle **240** may additionally or alternatively be positioned on a right side of hand-held power tool **100**. For example, a user may detach second handle **240** from a left side of hand-held power tool **100** and attach second handle **240** to a right side of hand-held power tool **100** as described above (e.g., using a fastener integral with housing **110**, etc.), thereby facilitating operation of hand-held power tool **100** by different individuals (e.g., right-handed individuals and left-handed individuals, etc.). In various embodiments, second handle **240** is positioned between motor **120** and air intake **150**, thereby facilitating a comfortable elbow position during operation.

In some embodiments, hand-held power tool **100** includes speed selector **250**. Speed selector **250** may facilitate selecting between a number of motor speeds. For example, a user may select a low speed for precise weeding and a high speed for digging through dirt. Speed selector **250** may control a rotational speed of motor **120**. In some embodiments, speed selector **250** may control a rotational direction of motor **120**. For example, speed selector **250** may facilitate switching between clockwise rotation of motor **120** and counterclockwise rotation of motor **120** (e.g., forward and reverse directions).

FIG. **5** illustrates a perspective view of exemplary trigger **210**. In some embodiments, trigger **210** includes switch **214**. Switch **214** may facilitate selecting a rotational direction of motor **120**. For example, switch **214** may facilitate switching between clockwise rotation of motor **120** and counterclockwise rotation of motor **120** (e.g., forward and reverse directions). Additionally or alternatively, switch **214** may facilitate locking trigger **210** so that it cannot be operated, thereby preventing accidental operation of hand-held power tool **100**. In some embodiments, switch **214** physically locks trigger **210** such that trigger **210** cannot be depressed. Additionally or alternatively, switch **214** may disable trigger **210** such that depression of trigger **210** no longer operates motor **120**. Switch **214** may be operated/toggled along axis **216**. For example, switch **214** may be positioned in a forward, reverse, and/or a lock position along axis **216**.

FIG. **6** illustrates a perspective view of exemplary weeding bit **600**. Weeding bit **600** is usable to remove plant material (e.g., weeds, etc.) from the ground. Weeding bit **600** is shown to include head **610** connected to connector **630** via shaft **620**. Head **610** may be mounted on an end of shaft **620** and may include one or more blades to facilitate removing plant material from the ground. Connector **630** may facilitate mechanically coupling weeding bit **600** to a driven shaft of motor **120**. For example, connector **630** may include a socket that slides over a driven shaft of motor **120**. In some embodiments, connector **630** includes coupling receiver **632** to receive a coupling mechanism (e.g., a bolt, a pin, etc.) to facilitate mechanically coupling weeding bit **600** to motor **120**.

Generally, as used herein, the term "substantially" is used to describe element(s) or quantitie(s) ideally having an exact quality (e.g., fixed, the same, uniformed, equal, similar, proportional), but practically having qualities functionally equivalent to the exact quality. For example, an element or

quantity is described as being substantially fixed or uniformed can deviate from the fixed or uniformed value, as long as the deviation is within a tolerance of the system (e.g., accuracy requirements, etc.). As another example, two elements or quantities described as being substantially equal can be approximately equal, as long as the difference is within a tolerance that does not functionally affect a system's operation.

Likewise, although some elements or quantities are described in an absolute sense without the term "substantially", it is understood that these elements and quantities can have qualities that are functionally equivalent to the absolute descriptions. For example, in some embodiments, a ratio is described as being one. However, it is understood that the ratio can be greater or less than one, as long as the ratio is within a tolerance of the system (e.g., accuracy requirements, etc.).

Although the disclosed embodiments have been fully described with reference to the accompanying drawings, it is to be noted that various changes and modifications will become apparent to those skilled in the art. Such changes and modifications are to be understood as being included within the scope of the disclosed embodiments as defined by the appended claims.

The terminology used in the description of the various described embodiments herein is for the purpose of describing particular embodiments only and is not intended to be limiting. As used in the description of the various described embodiments and the appended claims, the singular forms "a", "an," and "the" are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will also be understood that the term "and/or" as used herein refers to and encompasses any and all possible combinations of one or more of the associated listed items. It will be further understood that the terms "includes," "including," "comprises," and/or "comprising," when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

What is claimed is:

1. A hand-held power tool comprising:

a motor housing;

a motor in the motor housing, wherein the motor housing comprises an air intake spaced a distance from the motor;

a controller positioned between the air intake and the motor;

an auger coupled to the motor and comprising a rotational axis; and

a first handle coupled to the motor housing and including a grip, wherein the rotational axis of the auger and an axis of the grip subtend to an acute angle,

wherein the motor is positioned between the auger and a center of mass of the hand-held power tool.

2. The hand-held power tool of claim 1, wherein the acute angle is 65 to 90 degrees.

3. The hand-held power tool of claim 1, further comprising a clutch configured to interrupt transmission of torque from the motor to the auger when a threshold torque value is exceeded, wherein the clutch comprises a selector to select the threshold torque value from a plurality of torque values, and wherein the selector is mechanically limited to a single selection from the plurality of torque values.

4. The hand-held power tool of claim 1, wherein the acute angle is 74 degrees.

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5. The hand-held power tool of claim 1, further comprising a battery foot coupled to the first handle and configured to slidably engage a battery, wherein an axis of engagement of the battery foot and the rotational axis of the auger subtend a second angle of 15 to 75 degrees.

6. The hand-held power tool of claim 5, wherein the battery foot does not extend rearward of the axis of the grip.

7. The hand-held power tool of claim 5, further comprising a bridge coupled to the motor housing and the battery foot.

8. The hand-held power tool of claim 7, wherein the motor is positioned in the motor housing such that an axis of the bridge intersects the motor.

9. The hand-held power tool of claim 1, further comprising a second handle, wherein a longitudinal axis of the second handle is perpendicular to the rotational axis of the auger and the axis of the grip-portion.

10. A hand-held power tool comprising:

a motor housing;

a motor in the motor housing, wherein the motor housing comprises an air intake spaced a distance from the motor;

a controller positioned between the air intake and the motor;

a tool head coupled to the motor and comprising a rotational axis;

a battery foot configured to slidably engage a battery; and
a handle coupled to the motor housing and including a grip, wherein an axis of engagement of the battery foot and the rotational axis of the tool head subtend an angle greater than 0 degrees, and

wherein the motor is positioned between the tool head and a center of mass of the hand-held power tool.

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11. The hand-held power tool of claim 10, wherein the angle is 15 to 75 degrees.

12. The hand-held power tool of claim 10, further comprising a bridge coupled to the motor housing and the battery foot.

13. The hand-held power tool of claim 12, wherein the motor is positioned in the motor housing such that an axis of the bridge intersects the motor.

14. The hand-held power tool of claim 12, further comprising a second handle, wherein the second handle is positioned behind the motor.

15. The hand-held power tool of claim 10, wherein the motor is positioned proximal to the tool head.

16. The hand-held power tool of claim 10, wherein the tool head comprises an auger bit or a weeding bit.

17. A hand-held power tool comprising:

a motor housing;

a motor in the motor housing, wherein the motor housing comprises an air intake spaced a distance from the motor;

a controller positioned between the air intake and the motor;

a tool head coupled to the motor and comprising a rotational axis; and

a handle coupled to the motor housing and including a grip; and

wherein the motor is positioned between the tool head and a center of mass of the hand-held power tool.

18. The hand-held power tool of claim 17, further comprising a bridge coupled to the motor housing, wherein the motor is positioned in the motor housing such that an axis of the bridge intersects the motor.

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